AeroBook: 8–10 Minute Demo Video Script

Complete Dual-Track Presentation Script: Screen Actions & Natural English Voiceover

DURATION: 8 - 10 MINUTES | ARCHITECTURE: 11 MICROSERVICES | TARGET: PROFESSIONAL PRESENTATION | DUAL-TRACK: ACTION + VOICEOVER

PRE-RECORDING SETUP CHECKLIST ESSENTIAL

#	TARGET WINDOW / ITEM	URL OR ACTION	PURPOSE / ACCOUNT DETAILS
1	**Browser Tab 1 (Main UI)**	`http://localhost:4200`	Angular 18 Frontend Homepage (Landing Page)
2	**Browser Tab 2 (Discovery)**	`http://localhost:8761`	Netflix Eureka Dashboard showing all 10 services active
3	**Browser Tab 3 (Broker)**	`http://localhost:15672`	RabbitMQ Management UI (Seat & Notification queues)
4	**Passenger Account**	`priya@gmail.com`	Password: `Passenger@123` (Role: ROLE_USER)
5	**Staff Account**	`staff@aerobook.com`	Password: `Staff@123` (Role: ROLE_STAFF)
6	**Admin Account**	`admin@aerobook.com`	Password: `Admin@123` (Role: ROLE_ADMIN)
7	**Coupon Code for Demo**	`WELCOME10` or `AERO2026`	10% Instant Fare Discount Verification
8	**Display Resolution**	1920 × 1080 (1080p Full HD)	Zoom 100% or 110% for crystal-clear readability

PART 1: ARCHITECTURE, INNOVATION & FEATURES (0:00 – 4:00)

THEORY & CONTEXT

1. Introduction & Real-World Aviation Problem Statement

00:00 - 00:45

SCREEN ACTION

Display **AeroBook Landing Page** (`http://localhost:4200`). Hover smoothly over the flight search bar, date pickers, and top navigation bar.

VOICEOVER NARRATION

"Hello everyone, and welcome. Today, I am excited to demonstrate **AeroBook** — an enterprise-grade, cloud-native airline flight reservation, passenger services, airport ground operations, and fleet management platform.

In commercial aviation, legacy monolithic booking engines struggle with single points of failure, slow holiday booking spikes, and fragmented airport operations. If an email notification service slows down or a reporting query runs, it should never prevent a passenger from booking a flight or boarding an aircraft.

AeroBook was engineered from the ground up to solve these mission-critical challenges using a distributed, event-driven microservices architecture that guarantees fault isolation, high availability, sub-millisecond perimeter security, and real-time operational consistency."

2. Distributed Architecture & 11 Spring Boot Microservices

`00:45 - 01:45`

SCREEN ACTION

Switch tab to **Netflix Eureka Dashboard** (`http://localhost:8761`). Highlight registered instances: `CONFIG-SERVER` , `AUTH-SERVICE` , `API-GATEWAY` , `USER-SERVICE` , `PAYMENT-SERVICE` , `CHECK-IN-SERVICE` , `FLIGHT-SERVICE` , `BOOKING-SERVICE` , `FARE-SERVICE` , `NOTIFICATION-SERVICE` .

VOICEOVER NARRATION

"Under the hood, AeroBook is powered by **11 backend Java Spring Boot microservices** and an **Angular 18 Single Page Application**.

As you can see right here on our Eureka Service Discovery dashboard: At the perimeter, we have our **Spring Cloud API Gateway** running on Netty at port 8083, which acts as the single reverse proxy and security barrier for all incoming client traffic. We have centralized configuration management using **Spring Cloud Config Server**. The business tier is decomposed into independent domain services: `auth-service` for cryptographic JWT security, `user-service` for traveler profiles, `flight-service` for flight schedules and fleet management, `fare-service` for tiered pricing and coupons, `booking-service` for reservations, `payment-service` for transactions, `check-in-service` for airport boarding passes, and `notification-service` for email delivery.

Crucially, we enforce the **Database-per-Service pattern**. AeroBook runs **8 isolated MySQL database schemas**, guaranteeing zero cross-database locking between business domains."

3. Core Technical Innovations: Concurrency, AMQP & Perimeter Auth

`01:45 - 02:45`

SCREEN ACTION

Switch to **RabbitMQ Management** (`http://localhost:15672`) or display the architecture diagram slide showing asynchronous queues and Netty gateway routing.

VOICEOVER NARRATION

"Building a distributed platform like AeroBook involves resolving real distributed systems challenges. Here are three key innovations we engineered into the platform:

1. High-Concurrency Seat Locking & Anti-Overselling: To prevent race conditions when multiple users book the last available seat simultaneously, our Flight Service implements **pessimistic row-level locking** using `SELECT FOR UPDATE` within an isolated transactional boundary. This completely eliminates seat overselling.

2. Hybrid Eventual Consistency with RabbitMQ: Instead of slow, fragile distributed two-phase commits (2PC), we leverage **RabbitMQ AMQP messaging**. When a payment succeeds, the booking service asynchronously publishes seat deduction and e-ticket generation events, keeping microservice response times under 50 milliseconds.

3. Perimeter Authentication with Zero Internal Overhead: The Netty API Gateway decrypts and verifies JWT tokens at the boundary, and mutates downstream request headers with trusted identity claims (`X-User-Id` and `X-User-Role`). Downstream microservices evaluate roles instantly without repeated asymmetric cryptographic decodes."

4. Multi-Persona Experience & Frontend Technology

`02:45 - 03:30`

SCREEN ACTION

Switch back to **Angular Frontend** (`http://localhost:4200`). Scroll smoothly across the homepage showcasing the UI with Tailwind CSS and Lucide icons.

VOICEOVER NARRATION

"AeroBook delivers a complete multi-persona experience tailored to three distinct roles: The **Passenger** enjoys real-time flight search, tiered cabin selection, dynamic coupon application, Razorpay payment processing, interactive seat selection, digital PDF e-tickets, and web check-in. The **Airport Ground Staff** accesses a dedicated airport operations desk for passenger manifests, PNR lookups, real-time boarding pass barcode scanning, flight dispatching, and incident logging. The **Airline Administrator** commands fleet aircraft inventory, route and flight schedule creation, fare matrix controls, and user directory management.

On the frontend, AeroBook is built with **Angular 18**, utilizing standalone signals, Tailwind CSS, and automated SonarQube static code quality validation with JaCoCo and LCOV coverage across the entire codebase."

5. Transition to Live Demonstration

`03:30 - 04:00`

SCREEN ACTION

Move cursor smoothly to hover over the **Login** button in the top navigation bar.

VOICEOVER NARRATION

"Now that we have covered the business domain and the distributed architecture, let's jump straight into the **live demonstration**.

We will walk through the entire lifecycle: from a passenger booking a flight, applying a discount, and checking in, to ground staff boarding the passenger at the departure gate, and finally the administrative control center."

PART 2: LIVE END-TO-END SYSTEM WALKTHROUGH (4:00 – 10:00)

LIVE EXECUTION

Step 1: Passenger Login, Flight Search & Dynamic Fare Calculation

04:00 - 05:30

SCREEN ACTION

- Click **Login** and log in as `priya@gmail.com` / `Passenger@123`.
- Arrive on Customer Dashboard (`/customer`).
- Search Flight: Origin **DEL (Delhi)**, Destination **BOM (Mumbai)**, Date: Tomorrow, Class: Economy.
- Click **Search Flights**. Point out flight `AB-101` and available seats counter.
- Click **Select Flight**. Enter passenger name, age, gender.
- In the promo box, enter `WELCOME10` and click **Apply**. Point out the live discount deduction in the fare summary card.

VOICEOVER NARRATION

"We begin by logging in as our passenger, Priya. Notice how the API Gateway instantly authenticates our JWT and routes us directly to our dedicated Customer Dashboard.

Let's search for a flight from Delhi to Mumbai. The frontend queries our API Gateway, which load-balances the request through Eureka to the Flight Service.

Here are our live scheduled flights. Notice the real-time available seat counter. Let's select flight `AB-101`.

On the booking configuration screen, our Fare Service dynamically calculates the base fare, Passenger Service Fee (PSF), User Development Fee (UDF), and GST taxes.

Let's enter our passenger details, and now let's apply our promo code: `WELCOME10`. Instantly, the Fare Service verifies the coupon validity, applies the discount, and recalculates our final payable total."

Step 2: Razorpay Payment, E-Ticket Generation & Asynchronous AMQP

`05:30 - 06:45`

SCREEN ACTION

- Click **Proceed to Payment**. Razorpay Modal opens.
- Simulate successful payment and click **Pay Now**.
- Arrive on Confirmation Screen. Highlight the generated 8-character **PNR** (e.g., `ABK79421`).
- Click **View E-Ticket** to show the high-fidelity modal.
- Click **Download PDF** to show the OpenPDF official ticket with itinerary and tax breakdown.

VOICEOVER NARRATION

"Now let's proceed to payment. AeroBook integrates seamlessly with Razorpay.

When I click 'Pay', our backend payment service executes cryptographic HMAC-SHA256 signature verification to guarantee transaction authenticity and prevent tampering.

And just like that, the payment is confirmed! Behind the scenes, the Booking Service created a unique 8-character PNR, published an asynchronous event to RabbitMQ, and the Flight Service decremented the seat count in MySQL under row-level lock.

Simultaneously, the Notification Service dispatched an email confirmation with an attached PDF e-ticket. When I click 'View E-Ticket', we get this beautiful, print-ready document generated using OpenPDF, complete with the flight itinerary, fare breakdown, and official airline header."

Step 3: Web Check-In, Interactive Seat Map & Boarding Pass

`06:45 - 07:45`

SCREEN ACTION

- Click **Web Check-In**. Enter the PNR or select the booking.
- Aircraft fuselage seat map loads. Show occupied vs. available seats.
- Select window seat `12A`. Click **Confirm Check-In**.
- Show digital **Boarding Pass** with IATA barcode format (`BP-X-YYZ`), gate, and seat. Click **Download Boarding Pass**.

VOICEOVER NARRATION

"Now let's complete Web Check-In. Passengers can check in directly from their phone or browser. When I enter our PNR, AeroBook loads an interactive seat map representing the actual aircraft fuselage.

You can see occupied seats disabled in real time. Let's select seat `12A` by the window.

When I click 'Confirm Check-In', our `check-in-service` locks in the seat assignment, updates our status from `BOOKED` to `CHECKED_IN`, and generates our digital IATA-compliant Boarding Pass.

Notice the generated high-resolution barcode, terminal gate, boarding group, and sequence number. The passenger is now 100% ready for travel."

Step 4: Airport Ground Operations (Staff Portal & Gate Scanner) `07:45 - 08:45`

SCREEN ACTION

- Log out and log in as Staff: `staff@aerobook.com` / `Staff@123`.
- Arrive on Staff Dashboard (`/staff`).
- In Passenger Manifest, search for flight `AB-101` or PNR. Show Priya's record with status `CHECKED_IN`.
- Click **Simulate Gate Scan / Board Passenger**. Status updates to `BOARDED`.
- In Flight Status Dispatcher, update flight status from `SCHEDULED` to `BOARDING`.

VOICEOVER NARRATION

"Now let's switch personas to the airport terminal. I'll log in as airport ground staff using `staff@aerobook.com`.

Notice that our role-based security immediately recognizes `ROLE_STAFF` and directs us to the airport ground operations dashboard. Here, airport agents have real-time visibility across all flights, passenger manifests, and baggage counters.

Let's look up our flight `AB-101`. Here is our passenger, Priya, checked into seat `12A`.

At the departure gate, the agent scans the boarding pass barcode. With one click, the check-in service validates the ticket and marks the passenger as `BOARDED`.

The staff member can also update the operational flight status — let's switch flight `AB-101` from `SCHEDULED` to `BOARDING`. This status immediately propagates across the entire platform in real time."

Step 5: Airline Administration & Fleet Management

`08:45 - 09:30`

SCREEN ACTION

- Log out and log in as Admin: `admin@aerobook.com` / `Admin@123`.
- Arrive on Admin Dashboard (`/admin`).
- Click through tabs: **Fleet Management** (Boeing 737, Airbus A320), **Flight Schedules** (create/update schedules), **Coupons & Campaigns** (`WELCOME10`), and **User Management**.

VOICEOVER NARRATION

"Finally, let's log in as the System Administrator using `admin@aerobook.com`.

The Admin Control Center gives airline executives full command over the enterprise: Under **Fleet Management**, admins configure commercial aircraft, cabin seating layouts, and operational aircraft statuses. Under **Flight Schedules**, routes, departure/arrival timestamps, aircraft assignments, and base fares can be scheduled dynamically. Under **Coupons & Promotions**, marketing campaigns and discount codes can be created with usage limits and expiry dates. And under **User Management**, administrators can audit credentials, verify accounts, and manage role permissions across the entire company."

Conclusion & Technical Wrap-Up

`09:30 - 10:00`

SCREEN ACTION

Switch back to the **AeroBook Home Screen** or show the Eureka dashboard with all healthy registered services. Conclude smoothly.

VOICEOVER NARRATION

"To summarize: AeroBook demonstrates how modern cloud-native architectures solve real-world aviation challenges: **11 decoupled Spring Boot microservices** communicating via Netflix Eureka and Netty API Gateway. **Event-driven eventual consistency** powered by RabbitMQ. **Zero overselling** through row-level pessimistic locking. **Enterprise security** with JWT perimeter validation and BCrypt hashing. And a responsive, accessible **Angular 18 frontend** catering to Passengers, Staff, and Administrators alike.

Thank you very much for watching, and I am now open to any questions!"